

BRCA — Predict Molecular Subtypes from Gene Expression

Goal: Predict breast cancer PAM50 molecular subtypes using RNA-seq gene expression data, then integrate clinical metadata to deepen the biological interpretation.

Part I — Gene Expression Analysis

Step 1 — Data Structure Investigation

- How many samples and genes are in the expression matrix?
- How many annotation columns are retained after cleaning to `paper_columns`?
- What is the sample count per PAM50 subtype (`paper_BRCA_Subtype_PAM50`)?

Step 2 — Expression Distribution Analysis

- Randomly select 10 genes and visualize their expression distributions across all samples.
- Select the top 5 and bottom 5 most variable genes (by standard deviation); plot their distributions and compare.

Step 3 — Correlation Analysis

- Compute pairwise Pearson correlations across all samples using the full gene set.
- Visualize as a correlation heatmap. What structure do you observe? Are there sub-blocks?
- Interpret: does the correlation structure correspond to PAM50 subtypes?

Step 4 — Hierarchical Clustering

- Cluster samples using `hclust`.
- Cut the dendrogram to obtain `k` clusters (choose `k`; justify your choice).

- Compare cluster membership with PAM50 subtypes.

Step 5 — PCA Visualization

- Run prcomp on the transposed expression matrix (samples as rows).
- Plot PC1 vs. PC2 and PC1 vs. PC3, colored by PAM50 subtype and by hierarchical cluster.
- Report the variance explained by the first three PCs. What distinguishes the subtypes?

Step 6 — Differential Expression

- Use a Wilcoxon rank-sum test (one-vs-rest) to identify marker genes for each PAM50 subtype.
- Apply FDR correction (BH). Report the top 20 genes per subtype by adjusted p-value and log2 fold-change.
- Generate an annotated heatmap (rows = top DEGs, columns = samples, annotation bars = PAM50 subtype and pathologic stage).

Step 7 — Prediction Models

- Split data 70/30 (stratified by subtype).
- Binary task — tumor vs. normal: Logistic Regression, Random Forest, k-NN. Report confusion matrices and AUC-ROC.
- Multiclass task — predict PAM50 subtype: Random Forest, Multinomial Logistic Regression, k-NN.
- For each model, report: accuracy, F1, per-class precision/recall.

Step 8 — Method Comparison

- Compare model performance systematically in a summary table.
- Evaluate whether the top DEGs (Step 6) rank highly as predictive features (Step 7 feature importance). Overlap analysis with UpSet plot or Venn diagram.

Part II — Clinical Data Analysis

Step A — Annotation Cleaning

Tidy the annotation file to keep only curated, paper-level columns.

```
anot_clean <- brca_anot_clean[, c("barcode", grep("^paper_", names(brca_anot_clean),  
value = TRUE))]
```

Questions to address:

- How many paper_ columns are retained? What categories of information do they cover?
- Are there samples with missing subtype assignments? How should these be handled?

Step B — Clinical Variable Exploration

Explore the clinical metadata as a standalone dataset before merging with expression data.

- Explore paper_pathologic_stage, paper_BRCA_Pathology (IDC vs ILC), paper_age_at_initial_pathologic_diagnosis.
- Tabulate paper_vital_status and compute median follow-up time.
- Examine multi-omic cluster columns (paper_CNV.Clusters, paper_mRNA.Clusters, paper_Protein.Clusters) — how do they relate to PAM50?

Step C — Clinical Feature Classification

Predict a binary or multi-class clinical outcome using clinical variables only (no expression data).

3. Define the outcome variable and apply appropriate encoding.
4. Select a small set of meaningful features (age, stage, grade, subtype, etc.) and handle missing values by imputation or exclusion.
5. Train/test split (70/30 or 5-fold CV). Fit: Logistic Regression, Random Forest, k-NN.

6. Report confusion matrices, accuracy, sensitivity/specificity, and AUC-ROC for each model.
7. Identify the most predictive clinical features: variable importance from RF, and coefficients from Logistic Regression. Note that k-NN does not provide native feature importance.

Step D — Integrative Analysis (Expression + Clinical)

- Correlate top DEGs from Part I with continuous clinical variables (e.g., age) using Spearman's rho.
- Test whether top DEGs differ significantly across clinical stages.
- Build a multimodal classifier: combine top expression features (from Step 7 / RF importance) with key clinical features. Does adding clinical data improve performance?
- Visualize: boxplots of top expression features stratified by clinical subgroups; annotated heatmap with both molecular and clinical annotation bars.

Step E — Discussion & Interpretation

Critical thinking questions

Are the molecular subtypes (from expression clustering) concordant with clinical staging? Where do they disagree?

Which data modality — expression or clinical — gives better predictive accuracy for your outcome variable?

What biological mechanisms might explain the top DEGs found between clinical subgroups?

What are the limitations of the TCGA cohort for clinical prediction (sample size per stage, selection of academic-centre patients, etc.)?